

HW6 NEW SKlearn V2 - Filter / Wrapper / Embedded Feature Selection

This report restructures HW6 around the standard feature-selection framework: Filter Methods, Wrapper Methods, and Embedded Methods.

HW6 SKlearn V2 - Filter / Wrapper / Embedded Feature Selection

AI Engineer Project: feature-selection framework + regression/classification/PCA/KMeans pipeline

Rows

1000

Augmented dataset

Final features

R&D Spend, Marketing

Voting strategy

Best model

KNN Regressor

Baseline All Features

Best R2

0.989

RMSE 4,216

Clusters

KMeans=3

Silhouette 0.392

Feature Selection Framework

FILTER

Pearson + VIF + SelectKBest

Fast statistical screening before modeling.

WRAPPER

RFE

Model-driven recursive elimination.

EMBEDDED

LASSO + Random Forest Importance

Feature selection happens during training.

VOTING

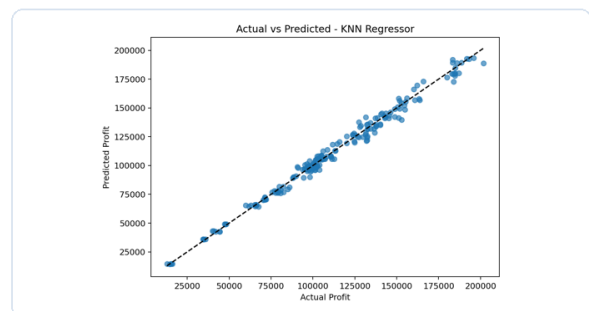
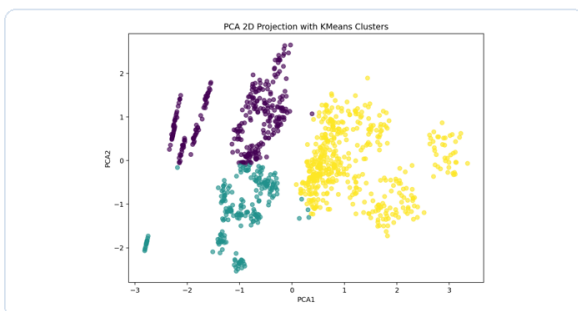
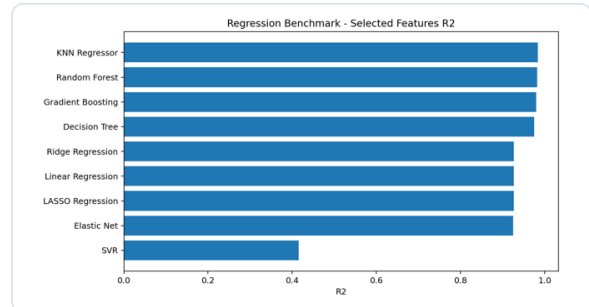
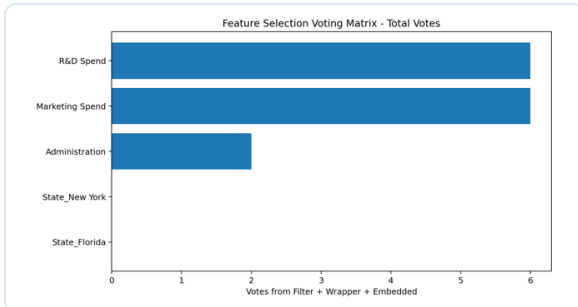
Total votes >= 3

Keep features supported by multiple schemes.

Feature voting decision matrix

Feature	Filter	Wrapper	Embedded	Votes	Decision
R&D Spend	3	1	2	6	Keep
Marketing Spend	3	1	2	6	Keep
Administration	1	0	1	2	Optional/Remove
State_New York	0	0	0	0	Optional/Remove
State_Florida	0	0	0	0	Optional/Remove

Technical evidence charts

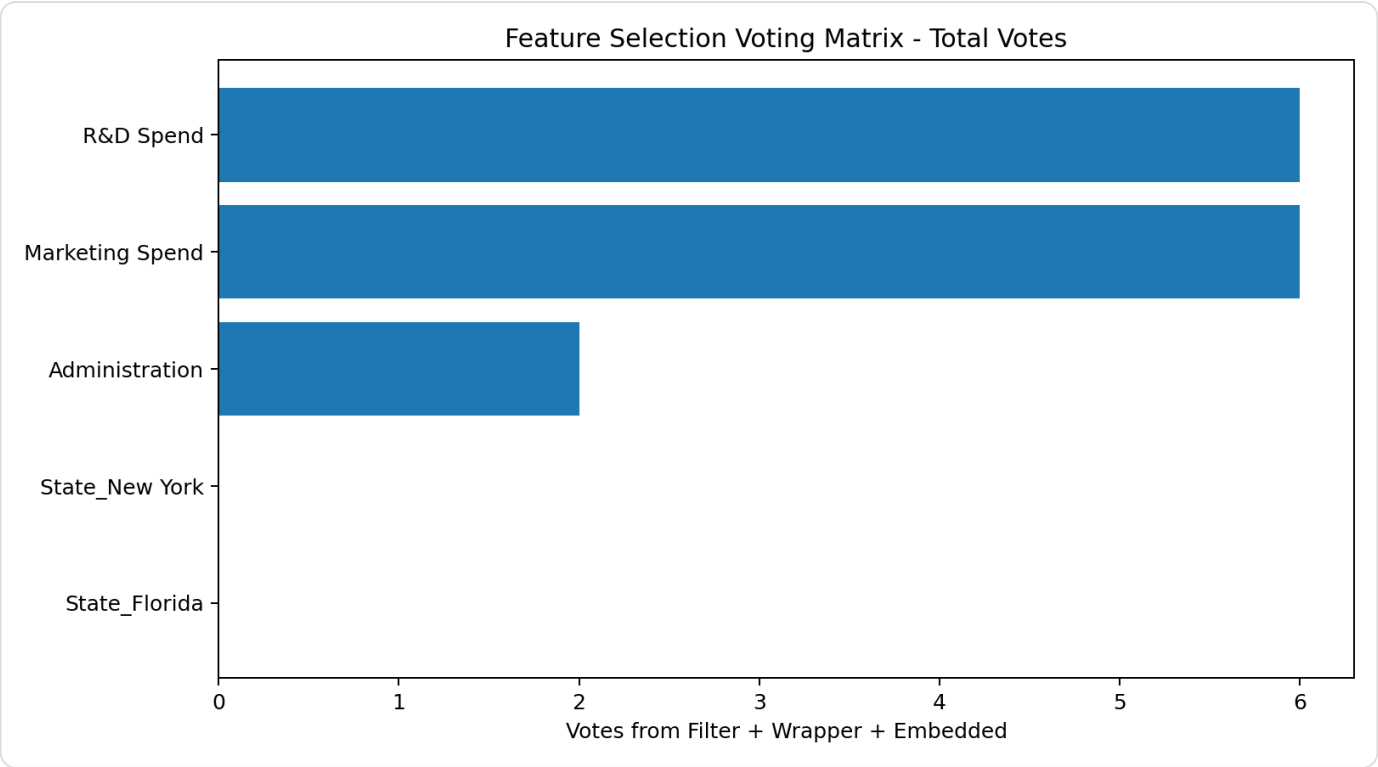


Final deployment design

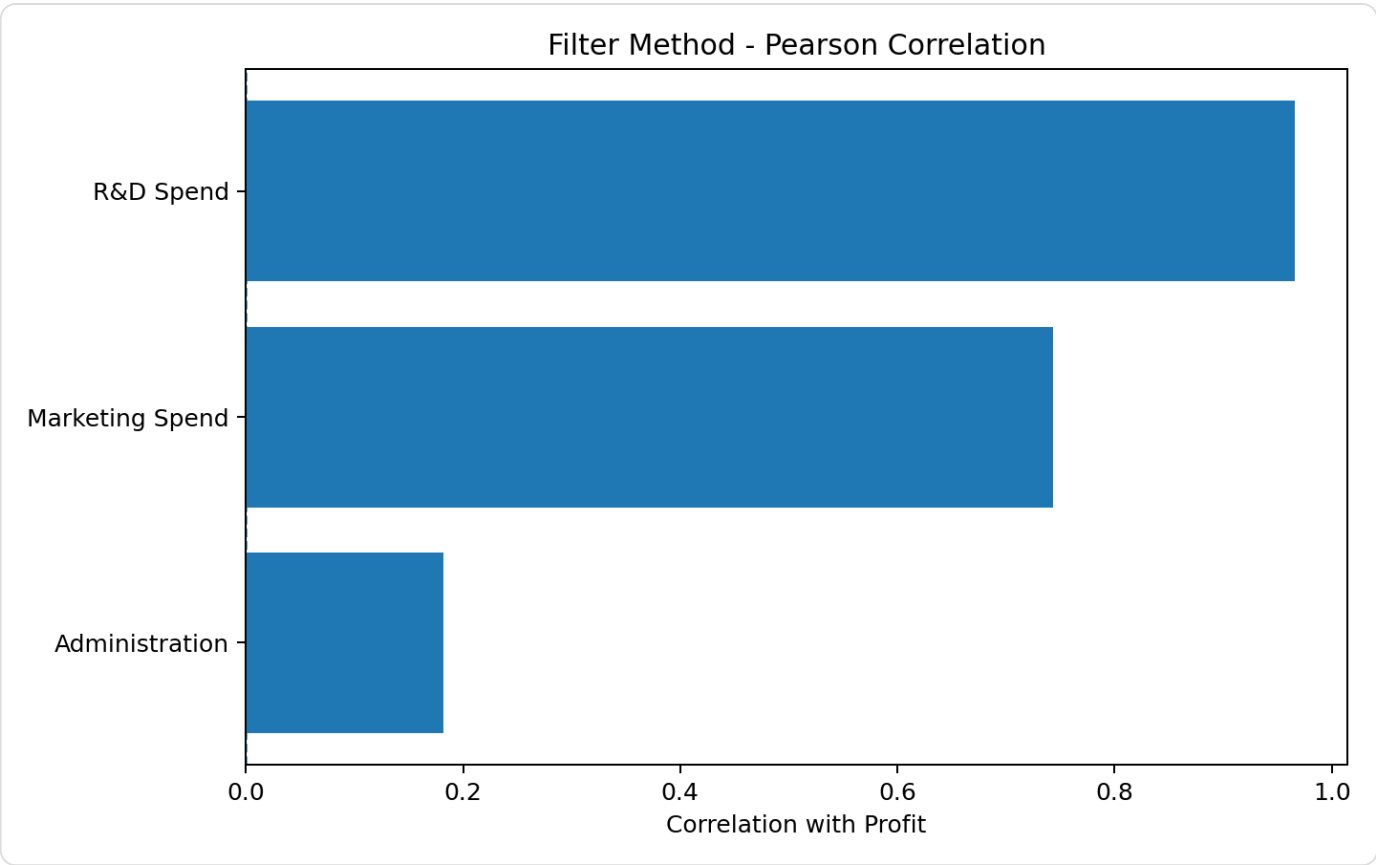
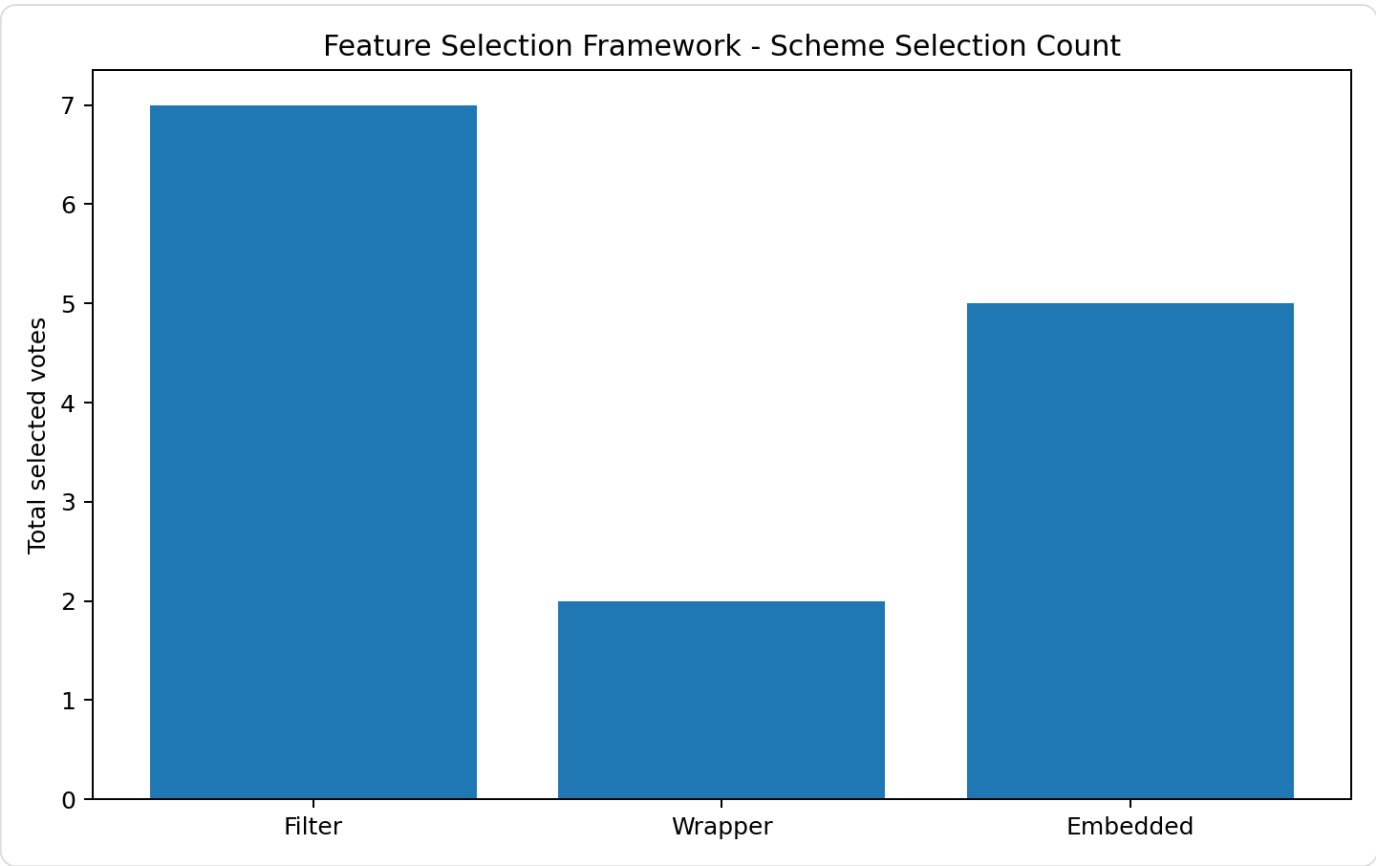
Input spending data -> StandardScaler / OneHotEncoder -> Best regression pipeline -> Predicted Profit; classifier pipeline -> Profit_Level; outputs saved as joblib.

Feature Selection Voting Matrix

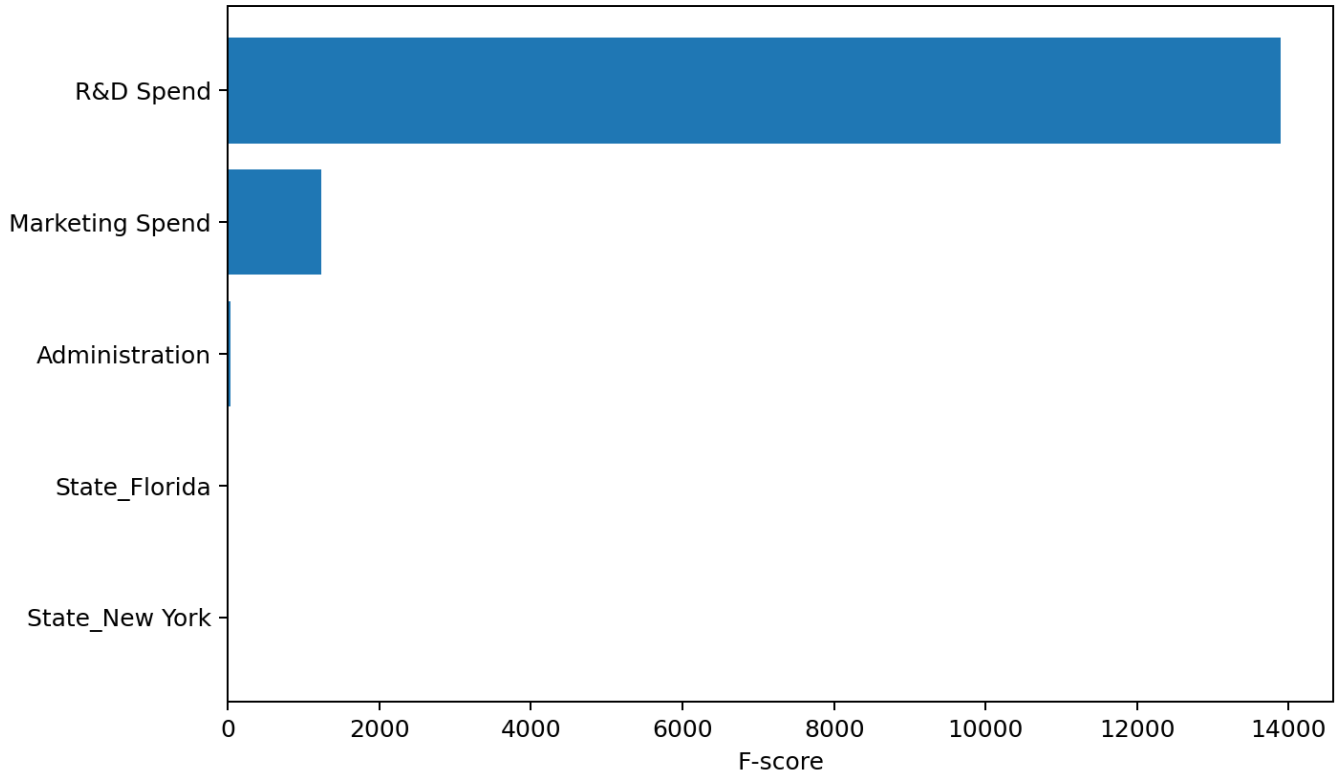
Feature	Pearson_Correlation	Filter_Pearson_Selected	VIF	Filter_VIF_Selected	F_score	Filter_SelectKBest_Selected	RFE_Rank	Wrapper
R&D Spend	0.965907	True	2.362117	True	13892.194439	True	1	True
Marketing Spend	0.743069	True	2.242695	True	1230.431857	True	1	True
Administration	0.182254	False	1.152308	True	34.289195	False	2	False
State_New York	NaN	False	NaN	False	0.993152	False	4	False
State_Florida	NaN	False	NaN	False	13.531090	False	3	False



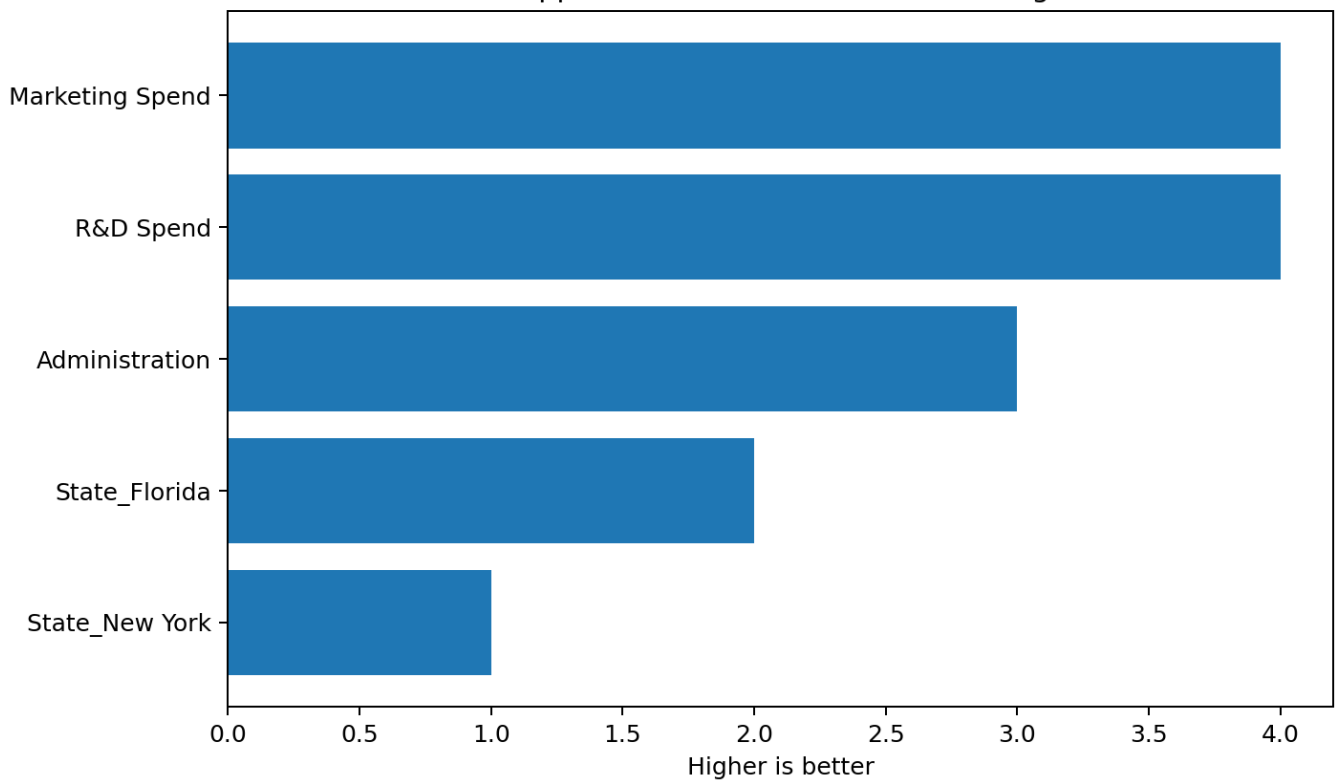
Feature Selection Framework Charts



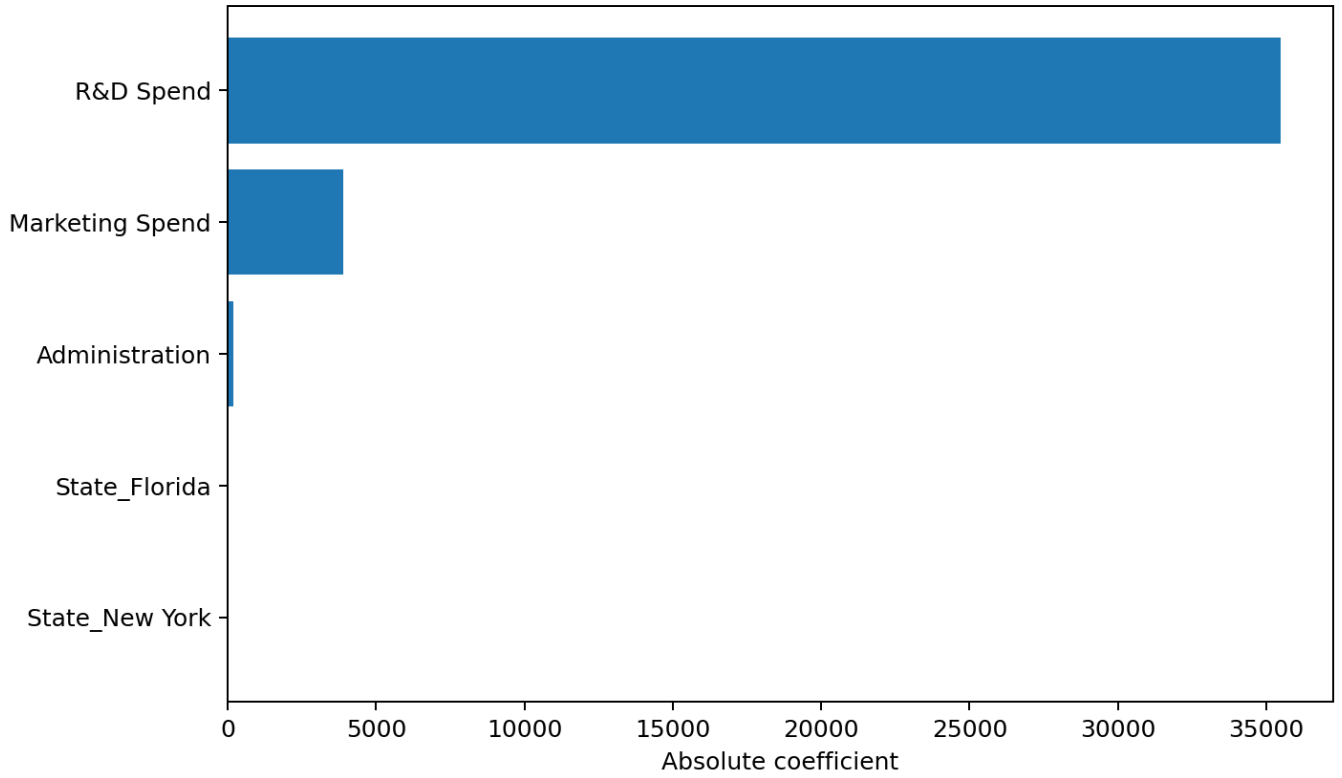
Filter Method - SelectKBest F-Regression



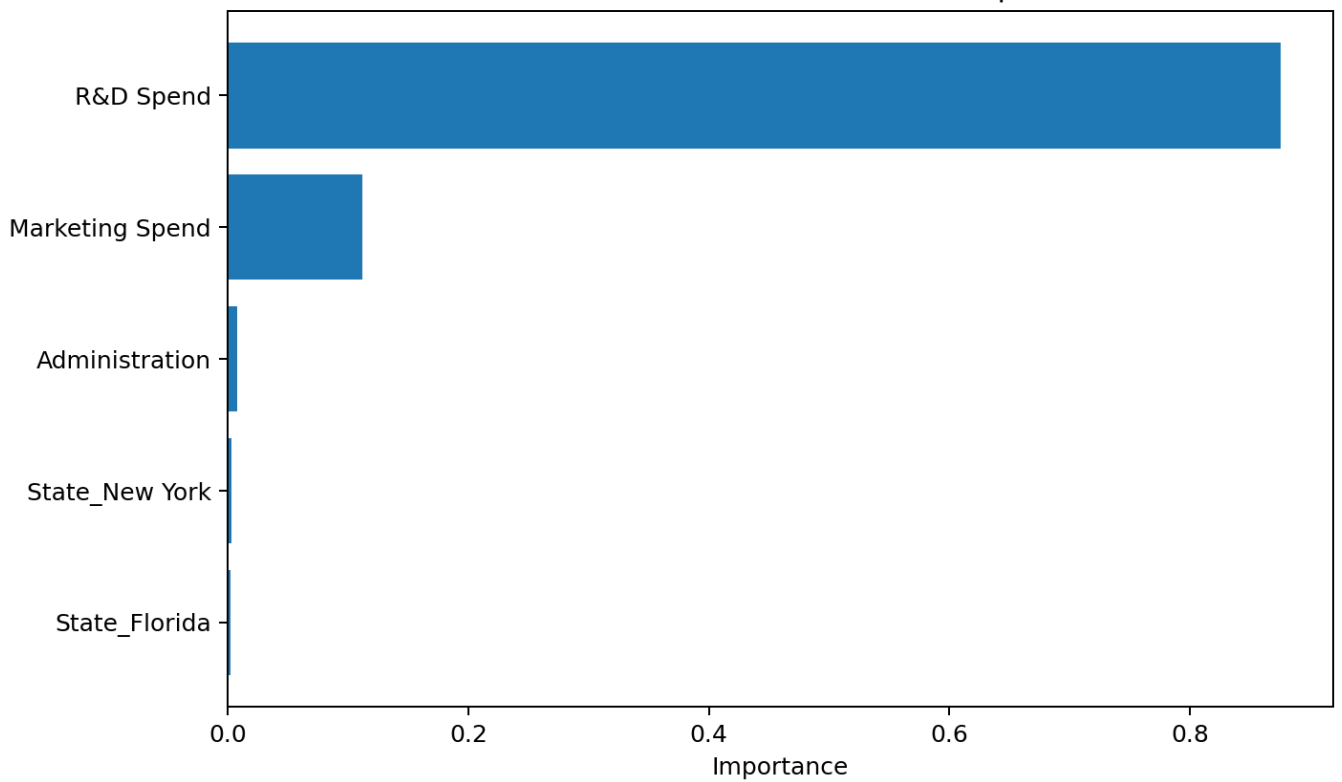
Wrapper Method - RFE Selection Strength



Embedded Method - LASSO Absolute Coefficients



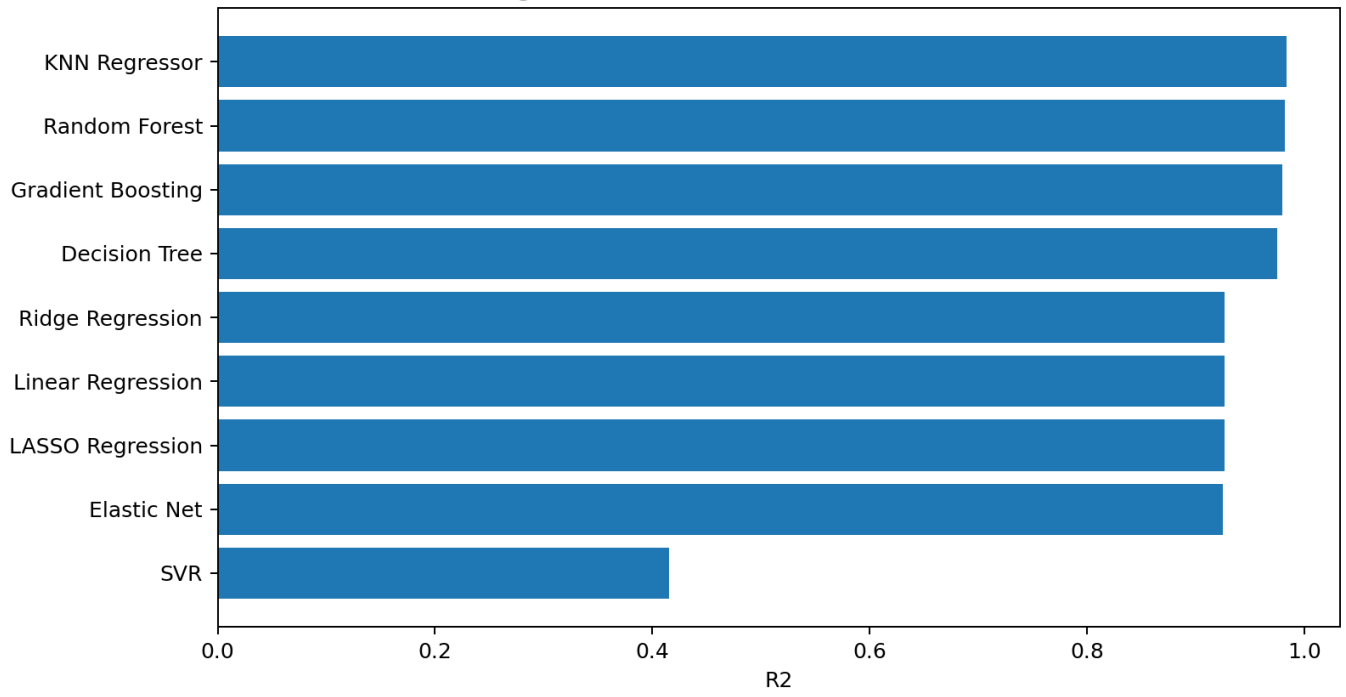
Embedded Method - Random Forest Importance



Regression Benchmark

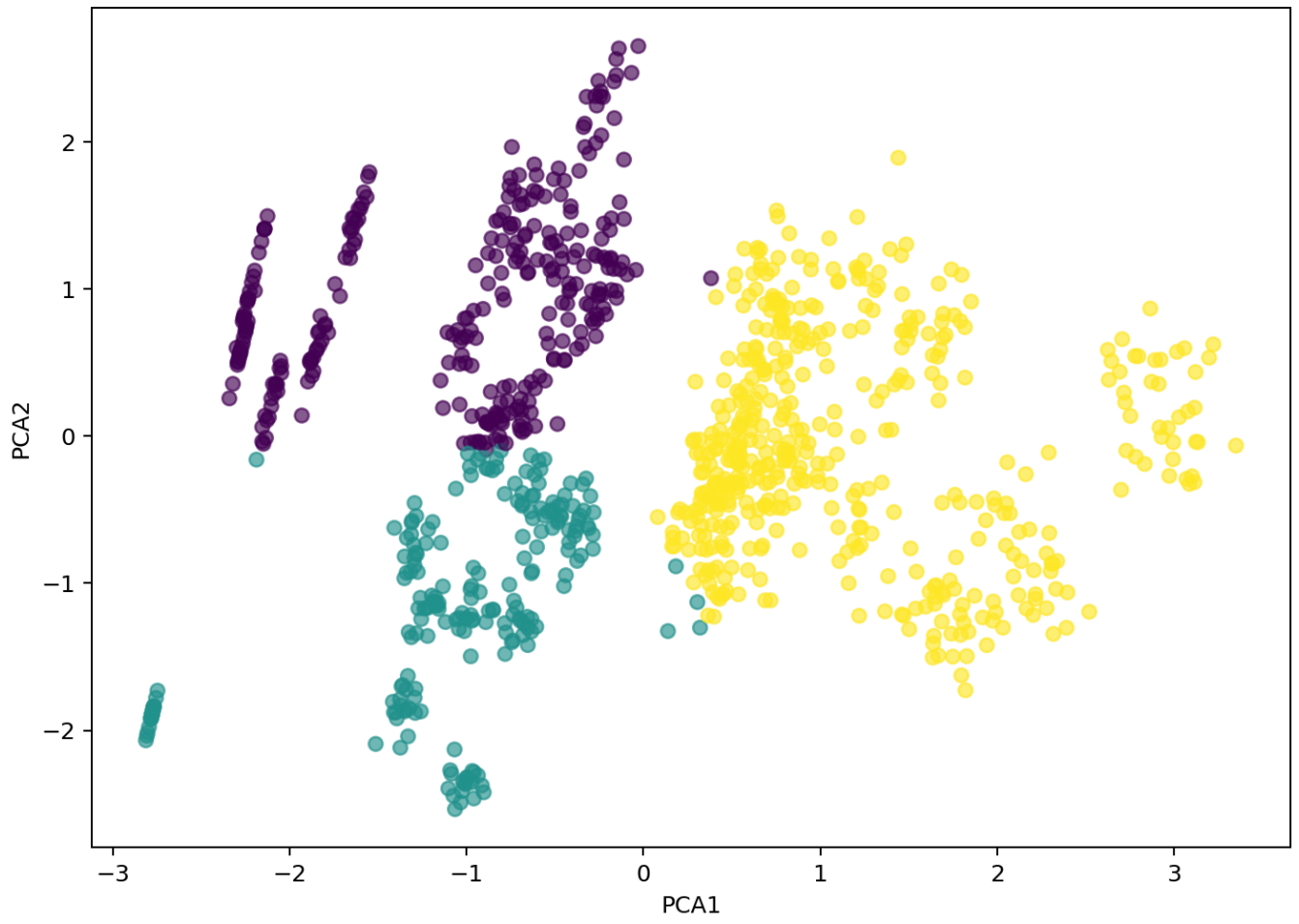
Feature_Set	Selected_Features	Model	MAE	RMSE	R2	CV_R2_Mean
Baseline_All_Features	R&D Spend, Administration, Marketing Spend, State	KNN Regressor	3272.323314	4216.393111	0.988945	0.989182
Baseline_All_Features	R&D Spend, Administration, Marketing Spend, State	Random Forest	3365.697665	4534.357795	0.987215	0.985644
Selected_Features_Filter_Wrapper_Embedded	R&D Spend, Marketing Spend	KNN Regressor	3789.412375	5127.524648	0.983651	0.984505
Selected_Features_Filter_Wrapper_Embedded	R&D Spend, Marketing Spend	Random Forest	3861.320695	5365.426522	0.982099	0.981233
Baseline_All_Features	R&D Spend, Administration, Marketing Spend, State	Gradient Boosting	4213.983433	5593.759789	0.980543	0.980656
Selected_Features_Filter_Wrapper_Embedded	R&D Spend, Marketing Spend	Gradient Boosting	4254.240909	5661.331258	0.980070	0.979180
Baseline_All_Features	R&D Spend, Administration, Marketing Spend, State	Decision Tree	3930.972359	5707.009546	0.979747	0.977938
Selected_Features_Filter_Wrapper_Embedded	R&D Spend, Marketing Spend	Decision Tree	4397.595738	6326.107543	0.975115	0.974280
Baseline_All_Features	R&D Spend, Administration, Marketing Spend, State	Linear Regression	8238.079196	10801.932850	0.927445	0.937348
Baseline_All_Features	R&D Spend, Administration, Marketing Spend, State	LASSO Regression	8238.091322	10801.940328	0.927445	0.937349
Baseline_All_Features	R&D Spend, Administration, Marketing Spend, State	Ridge Regression	8237.819234	10803.242325	0.927427	0.937347
Selected_Features_Filter_Wrapper_Embedded	R&D Spend, Marketing Spend	Ridge Regression	8307.698138	10854.216837	0.926741	0.937525
Selected_Features_Filter_Wrapper_Embedded	R&D Spend, Marketing Spend	Linear Regression	8311.802303	10854.864021	0.926732	0.937528
Selected_Features_Filter_Wrapper_Embedded	R&D Spend, Marketing Spend	LASSO Regression	8311.822897	10854.874766	0.926732	0.937528
Selected_Features_Filter_Wrapper_Embedded	R&D Spend, Marketing Spend	Elastic Net	8283.753764	11010.171984	0.924621	0.934835
Baseline_All_Features	R&D Spend, Administration, Marketing Spend, State	Elastic Net	8329.796587	11046.712417	0.924119	0.934336
Selected_Features_Filter_Wrapper_Embedded	R&D Spend, Marketing Spend	SVR	21423.561977	30664.425694	0.415298	0.394076
Baseline_All_Features	R&D Spend, Administration, Marketing Spend, State	SVR	24804.378123	33643.652429	0.296165	0.268959

Regression Benchmark - Selected Features R2

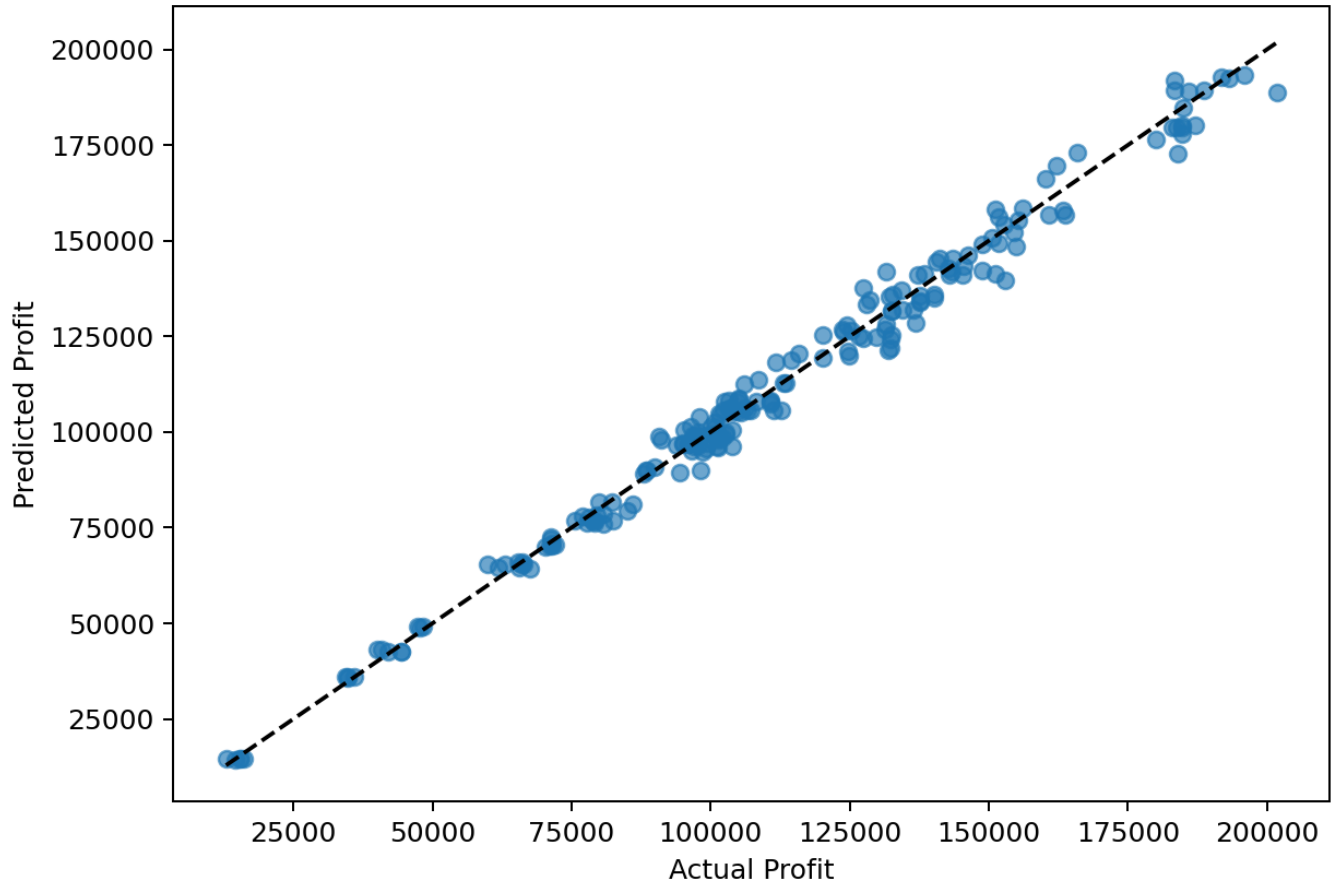


PCA and Actual vs Predicted

PCA 2D Projection with KMeans Clusters



Actual vs Predicted - KNN Regressor



Markdown Report

```
# HW6 NEW SKlearn V2 Technical Report
```

```
## Executive Summary
```

This version reorganizes the project around the most standard machine learning feature-selection framework:

```
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Filter Methods
Wrapper Methods
Embedded Methods
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```

The goal is to make the HW6 project more professional and easier to explain in a technical report or AI engineer portfolio.

```
## Dataset
```

```
Item	Value
Rows	1000
Numeric features	R&D Spend, Administration, Marketing Spend
Categorical feature	State
Regression target	Profit
Classification target	Profit_Level
```

```
## Feature Selection Framework
```

```
### 1. Filter Methods
```

Filter methods evaluate features before model training.

Used methods:

- Pearson Correlation
- VIF

- SelectKBest F-Regression

Purpose:

- Quickly identify direct feature-target relationships.
- Detect multicollinearity.
- Rank features statistically.

2. Wrapper Methods

Wrapper methods use a model to test feature subsets.

Used method:

- Recursive Feature Elimination (RFE)

Purpose:

- Iteratively remove weak features based on model behavior.
- Validate which features are useful when the model is actually trained.

3. Embedded Methods

Embedded methods perform feature selection during model training.

Used methods:

- LASSO Regression
- Random Forest Feature Importance

Purpose:

- LASSO shrinks weak coefficients toward zero.
- Random Forest captures nonlinear feature contribution.

Voting Strategy

The project combines the three groups into a voting matrix.

A feature receives votes from:

1. Pearson Correlation
2. VIF
3. SelectKBest
4. RFE
5. LASSO
6. Random Forest Importance

Final rule:

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Total votes >= 3 -> Keep

Total votes < 3 -> Optional / Remove

```
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```

Feature Voting Matrix

| Feature | Pearson_Correlation | Filter_Pearson_Selected | VIF | Filter_VIF_Selected | F_score | Filter_SelectKBest_Selected | RFE_Rank | Wrapper_RFE_Selected | LASSO_Coefficient | Embedded_LASSO_Selected | RF_Importance |
|----------------------|---------------------|-------------------------|----------------|---------------------|-----------------|-----------------------------|----------|----------------------|-------------------|-------------------------|---------------|
| Embedded_RF_Selected | Filter_Score | Wrapper_Score | Embedded_Score | Total_Votes | Decision | | | | | | |
| R&D Spend | 0.965907 | True | 2.36212 | True | 13892.2 | | | | | | |
| True | 1 | True | 35487.4 | True | | 0.875348 | | | | | |
| True | 3 | 1 | 2 | 6 | Keep | | | | | | |
| Marketing Spend | 0.743069 | True | 2.24269 | True | 1230.43 | | | | | | |
| True | 1 | True | 3882.5 | True | | 0.111512 | | | | | |
| True | 3 | 1 | 2 | 6 | Keep | | | | | | |
| Administration | 0.182254 | False | 1.15231 | True | 34.2892 | | | | | | |
| False | 2 | False | -194.53 | True | | 0.00742939 | | | | | |
| False | 1 | 0 | 1 | 2 | Optional/Remove | | | | | | |
| State_New York | nan | False | nan | False | 0.993152 | | | | | | |
| False | 4 | False | 0 | False | | 0.00314306 | | | | | |
| False | 0 | 0 | 0 | 0 | Optional/Remove | | | | | | |
| State_Florida | nan | False | nan | False | 13.5311 | | | | | | |
| False | 3 | False | -0 | False | | 0.00256717 | | | | | |
| False | 0 | 0 | 0 | 0 | Optional/Remove | | | | | | |

Final Selected Features

```
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```

R&D Spend, Marketing Spend

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```

Regression Benchmark

| Feature_Set | | Selected_Features | Model | MAE | RMSE |
|---|------------|---|-------------------|---------|---------|
| R2 | CV_R2_Mean | | | | |
| Baseline_All_Features | | R&D Spend, Administration, Marketing Spend, State | KNN Regressor | 3272.32 | 4216.39 |
| 0.988945 | 0.989182 | | | | |
| Baseline_All_Features | | R&D Spend, Administration, Marketing Spend, State | Random Forest | 3365.7 | 4534.36 |
| 0.987215 | 0.985644 | | | | |
| Selected_Features_Filter_Wrapper_Embedded | | R&D Spend, Marketing Spend | KNN Regressor | 3789.41 | 5127.52 |
| 0.983651 | 0.984505 | | | | |
| Selected_Features_Filter_Wrapper_Embedded | | R&D Spend, Marketing Spend | Random Forest | 3861.32 | 5365.43 |
| 0.982099 | 0.981233 | | | | |
| Baseline_All_Features | | R&D Spend, Administration, Marketing Spend, State | Gradient Boosting | 4213.98 | 5593.76 |
| 0.980543 | 0.980656 | | | | |
| Selected_Features_Filter_Wrapper_Embedded | | R&D Spend, Marketing Spend | Gradient Boosting | 4254.24 | 5661.33 |
| 0.98007 | 0.97918 | | | | |
| Baseline_All_Features | | R&D Spend, Administration, Marketing Spend, State | Decision Tree | 3930.97 | 5707.01 |
| 0.979747 | 0.977938 | | | | |
| Selected_Features_Filter_Wrapper_Embedded | | R&D Spend, Marketing Spend | Decision Tree | 4397.6 | 6326.11 |
| 0.975115 | 0.97428 | | | | |
| Baseline_All_Features | | R&D Spend, Administration, Marketing Spend, State | Linear Regression | 8238.08 | 10801.9 |
| 0.927445 | 0.937348 | | | | |
| Baseline_All_Features | | R&D Spend, Administration, Marketing Spend, State | LASSO Regression | 8238.09 | 10801.9 |
| 0.927445 | 0.937349 | | | | |
| Baseline_All_Features | | R&D Spend, Administration, Marketing Spend, State | Ridge Regression | 8237.82 | 10803.2 |
| 0.927427 | 0.937347 | | | | |
| Selected_Features_Filter_Wrapper_Embedded | | R&D Spend, Marketing Spend | Ridge Regression | 8307.7 | 10854.2 |
| 0.926741 | 0.937525 | | | | |
| Selected_Features_Filter_Wrapper_Embedded | | R&D Spend, Marketing Spend | Linear Regression | 8311.8 | 10854.9 |
| 0.926732 | 0.937528 | | | | |
| Selected_Features_Filter_Wrapper_Embedded | | R&D Spend, Marketing Spend | LASSO Regression | 8311.82 | 10854.9 |
| 0.926732 | 0.937528 | | | | |
| Selected_Features_Filter_Wrapper_Embedded | | R&D Spend, Marketing Spend | Elastic Net | 8283.75 | 11010.2 |
| 0.924621 | 0.934835 | | | | |
| Baseline_All_Features | | R&D Spend, Administration, Marketing Spend, State | Elastic Net | 8329.8 | 11046.7 |
| 0.924119 | 0.934336 | | | | |
| Selected_Features_Filter_Wrapper_Embedded | | R&D Spend, Marketing Spend | SVR | 21423.6 | 30664.4 |
| 0.415298 | 0.394076 | | | | |
| Baseline_All_Features | | R&D Spend, Administration, Marketing Spend, State | SVR | 24804.4 | 33643.7 |
| 0.296165 | 0.268959 | | | | |

Classification Benchmark

| Model | Accuracy | F1_Macro |
|--------------------------|----------|----------|
| Logistic Regression | 0.91 | 0.910813 |
| Random Forest Classifier | 0.905 | 0.905413 |

PCA and Clustering

```
Item	Value
PCA components	2
PCA explained variance	0.9207
KMeans clusters	3
Silhouette score	0.3917
```

Final AI Engineering Conclusion

Using the Filter / Wrapper / Embedded framework makes the feature-selection process more reliable and easier to justify.

The final recommendation is to use:

```
```text
R&D Spend + Marketing Spend
```
```

as the optimized regression feature set.

`Administration` and `State` can be kept for baseline comparison, but the voting matrix shows they are not essential for the optimized production model.

The project is ready for Streamlit or FastAPI deployment using the saved joblib pipelines.